
Neural Thickets: Diverse Task Experts Are Dense Around Pretrained Weights

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ICML 2026 Spotlight

- Weight-space analysis
- LLM post-training
- Gradient-free optimization

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Contents

Neural Thickets (IMCL 2026 Spotlight)

- 01 Introduction
- 02 Observations
- 03 Method
- 04 Experiments
- 05 Conclusion & Limitations

| purpose of study

- 기존 관점: 좋은 모델 파라미터는 찾기 어렵다. Gradient descent가 필요하다.
- 질문: 그런데 큰 사전학습 모델 주변에는 이미 좋은 해가 많지 않을까?
- 힘들게 건초더미에서 바늘을 찾지 말고 그 주변을 탐색하자

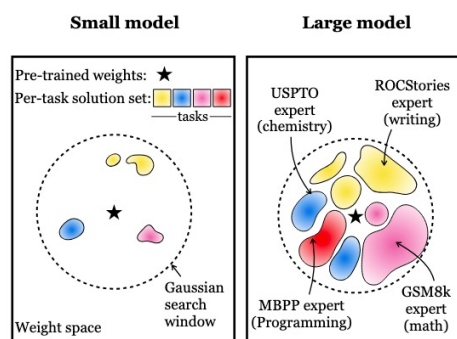
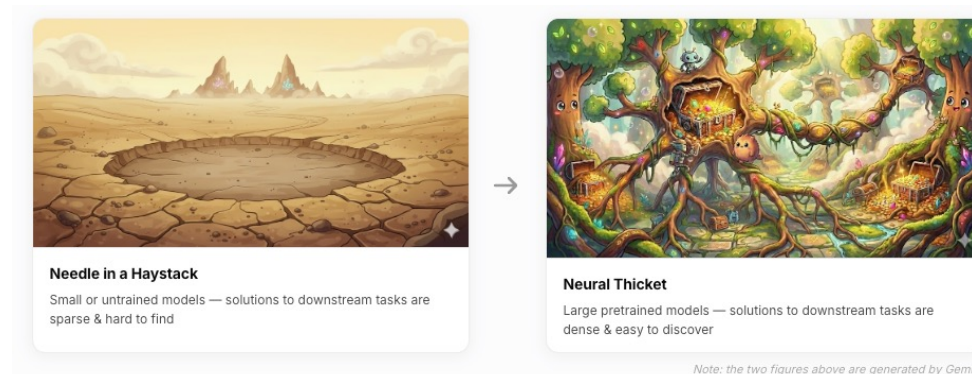


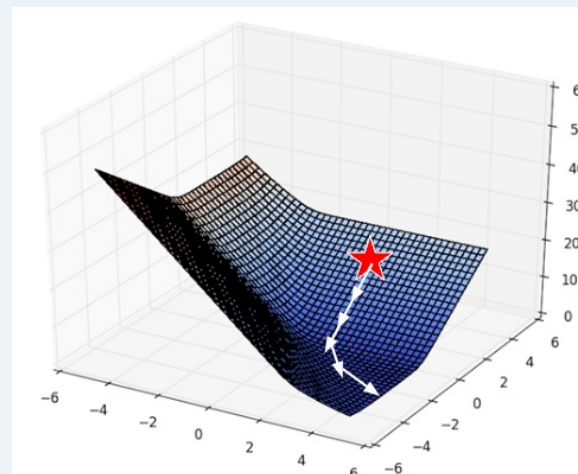
Figure 1(a)



1. **Density**
Larger pretrained models have more task-improving solutions nearby.
2. **Diversity**
These nearby solutions are different **task specialists**, not uniform generalists.
3. **Exploitability**
Randomly sampling and selecting these solutions can already provide competitive post-training.

AS-IS

- Structured Search
 - Gradient descent / fine-tuning
 - PPO
 - GRPO
 - Evolution Strategies (ES)
- **Implicit assumption:**
Good downstream solutions occupy only a **tiny** fraction of parameter space.
- Randomly guessing billions of parameters from scratch is essentially **impossible!!!!**



Gap: What does pretraining put around the solution?

| From a Needle to a Thicket

- Small / Untrained Models
 - Nearby good solutions are **rare**
 - Random perturbation usually hurts performance
 - **Structured optimization is necessary**
- Large, Well-Pretrained Models
 - Nearby task-improving solutions become **dense**
 - Different perturbations expose **different capabilities**
 - **Random sampling may become sufficient**

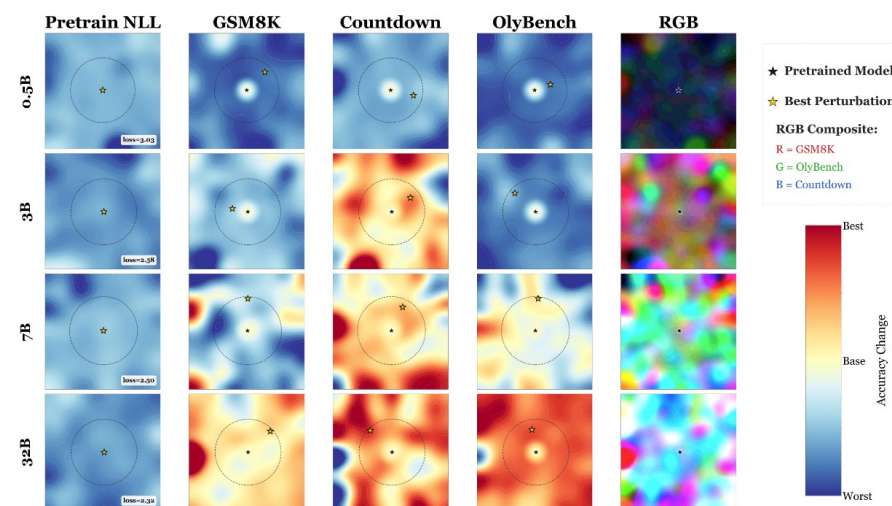


Figure 2: Accuracy Landscape

| Finding 1: Solution Density Scales with Model Size

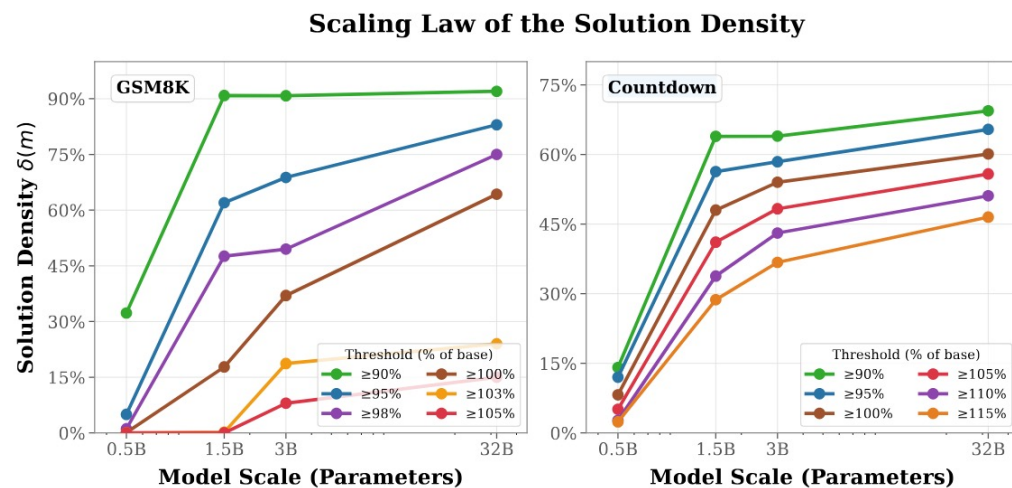


Figure 3(a): Solution Density Scaling

How to measure it?

$$\delta(m) = \mathbb{P}_{\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})} [s(\boldsymbol{\theta} + \epsilon) \geq s(\boldsymbol{\theta}) + m]$$

| Finding 2: Nearby Solutions Are Diverse Specialists, not Generalists

1. H1 - Generalists

- GSM8K ↑
- MATH ↑
- MBPP ↑
- USPTO ↑

2. H2 - Specialists

- Math ↑ ↑
- Coding ↓
- Chemistry ↓

Scaling Law of the Solution Diversity

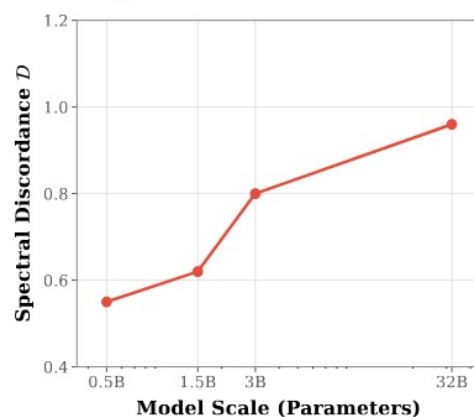


Figure 3(b)

How to measure it?

$$\mathcal{D} = 1 - \frac{1}{M(M-1)} \sum_{j \neq k} c_{jk}$$

- c_{jk} : task j 와 k 의 performance ranking 사이 상관계수
- $\mathcal{D} \rightarrow 0$: similar rankings → generalists
- $\mathcal{D} \rightarrow 1$: different rankings → specialists

| Finding 2: Nearby Solutions Are Diverse Specialists, not Generalists

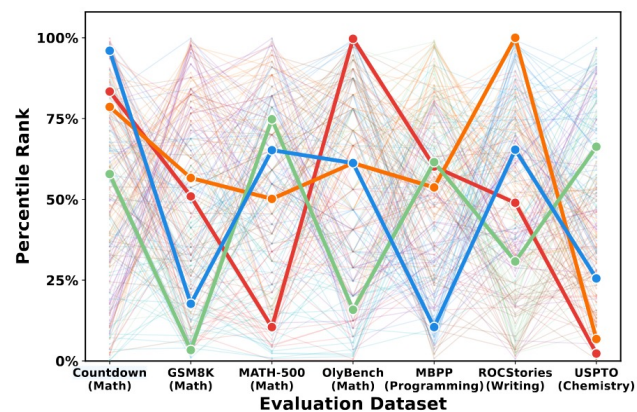
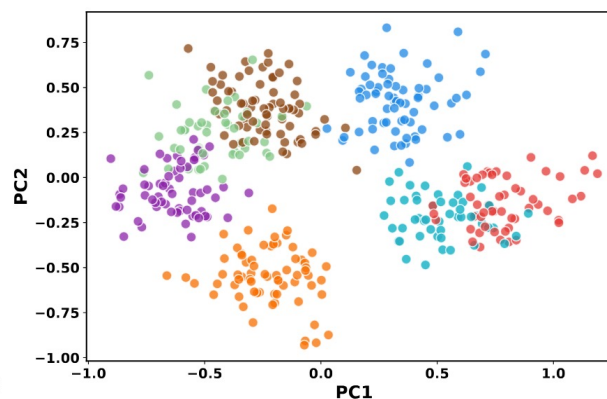


Figure 4(a)



- 500 perturbations
- 7 tasks
- 4 domains:
 - Math
 - Code
 - Writing
 - Chemistry

| Why Do Thickets Emerge?

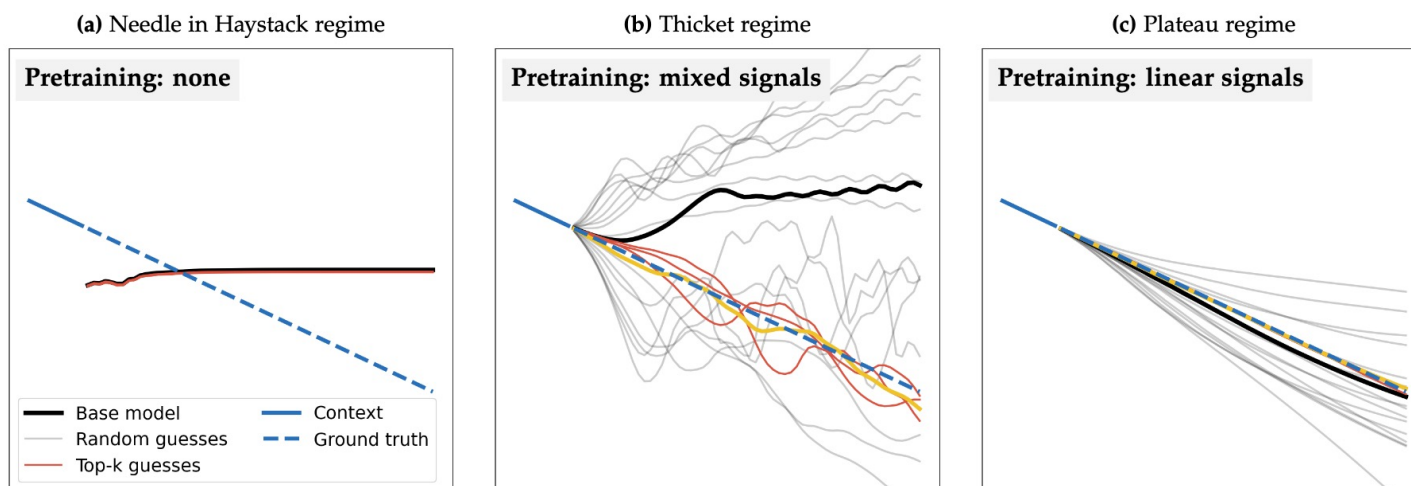


Figure 5: Autoregressive next-value prediction using MLP

| RandOpt: Random Guessing + Selection + Ensembling

Training – Random Guessing & Checking

1. Sample N random perturbation
2. Evaluate each model on D_{train}
3. Rank by validation/task performance
4. Select Top-K

Inference

For input x :

- Run selected K models
- Aggregate predictions with majority vote

$$\hat{y} = \text{mode} \left(\left\{ \arg \max_y f_{\theta_i}(y|x) \mid i \in \mathcal{I}_{\text{top}} \right\} \right)$$

What RandOpt does NOT do

- ✗ Gradient computation
- ✗ Parameter update trajectory
- ✗ Sequential optimization

What it does

- ✓ Fully parallel sampling
- ✓ Selection
- ✓ Ensemble of specialists

| Experimental Setup

LLMs

- Qwen
- Llama
- OLMo3
- 0.5B-8B

Tasks

Math

- Countdown
- GSM8K
- MATH-500
- OlympiadBench

Code

- MBPP

Writing

- ROCStoriesGSM8K

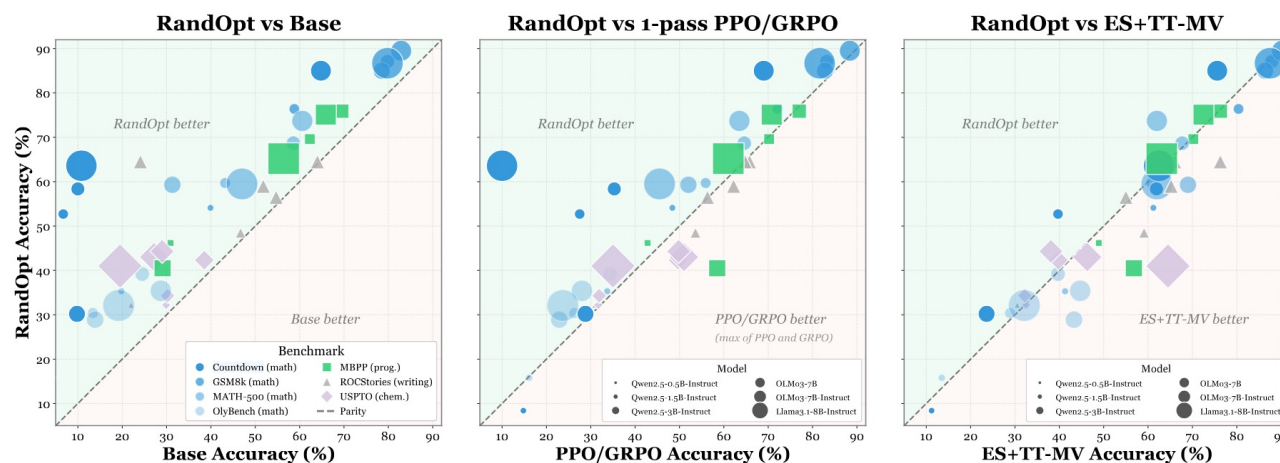
Chemistry

- USPTO

Baselines

- Base
- Test-Time Majority Vote (TT-MV)
- Best-of-N
- PPO
- GRPO
- Evolution Strategies (ES)
- Post-training methods are compared under equal training FLOPs !

| RandOpt Is Competitive with Standard Post-training



Important: RandOpt $K=50$ uses **test-time ensembling**, while standard PPO/GRPO results generally use one test-time sample !

| Rethinking Pretraining and Post-training

Three Takeaways

1. Density
 - Large, well-pretrained models에서는 task-improving solutions이 pretrained weights 주변에 조밀하게 존재한다.
2. Diversity
 - 그리고 그 solution들은 서로 다른 task-specific capabilities를 갖는다.
3. Post-training as Selection
 - Pretrained weights 주변에 이미 존재하는 task-specific solution을 탐색하고 선택하는 과정으로 볼 수 있다.

Limitations

- RandOpt는 대규모 sampling 과 Test-Time ensemble에 의존한다.
- Test-time cost는 선택된 experts K 의 수만큼 증가한다.
- 성능 향상이 반드시 새로운 reasoning capability의 학습을 의미하지는 않는다.

Pretraining may not only learn a solution — it may construct a neighborhood of solutions.